

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)	(i)		$75 \times 16 = 1200$ [MBq]		1		1	1	
		(ii)		Halving 1200 (ecf) or 16 at least once (1) 1200 (ecf) $\times \frac{1}{2} \times \frac{1}{2} = 300$ [MBq] (1)		2		2	2	
	(b)			Too short a half-life or decays too quickly or doesn't last long enough or pellet would need replacing (1) so wouldn't kill enough [cancer] <u>cells</u> (1)		2		2		
				Question 8 total	0	5	0	5	3	0